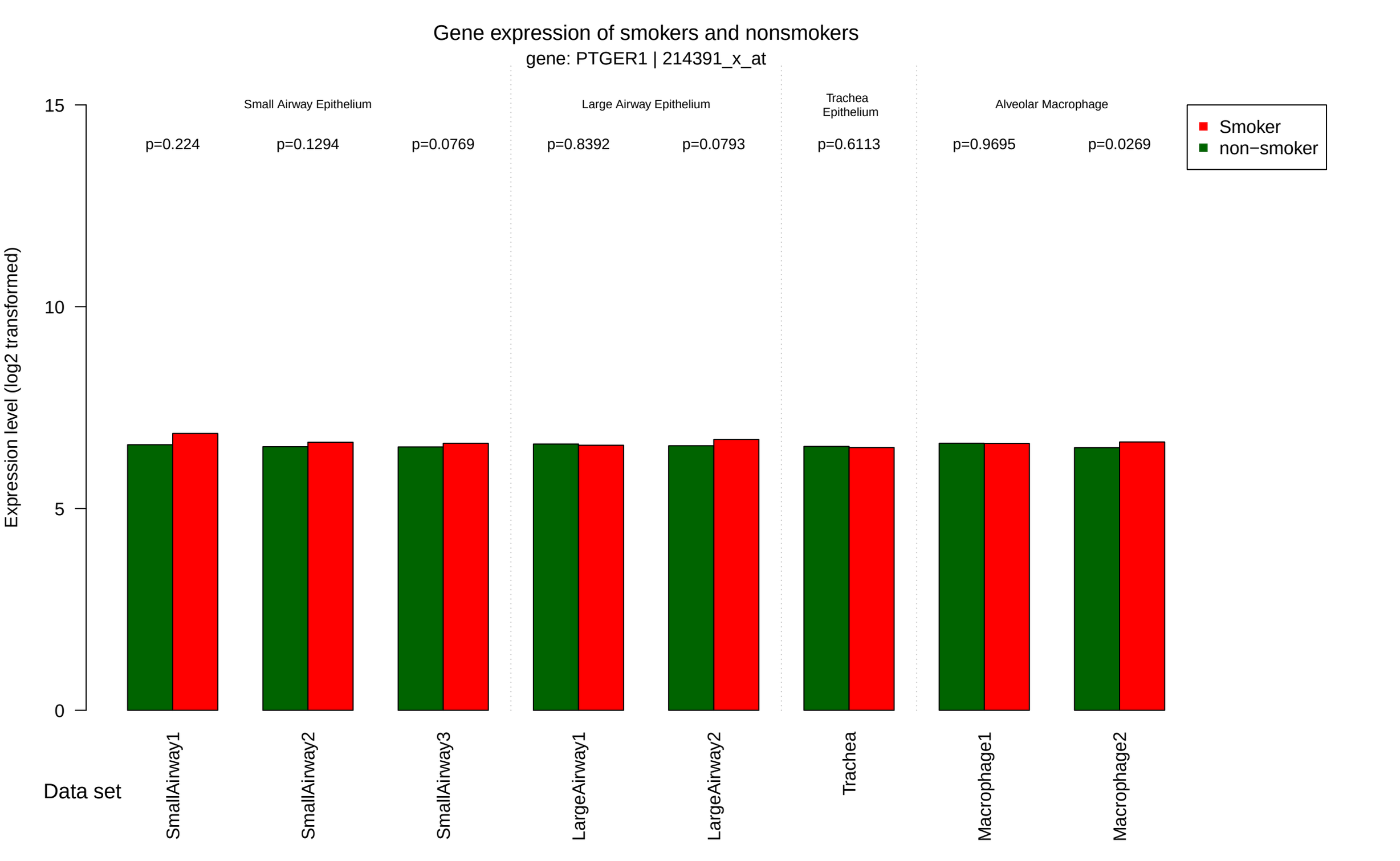
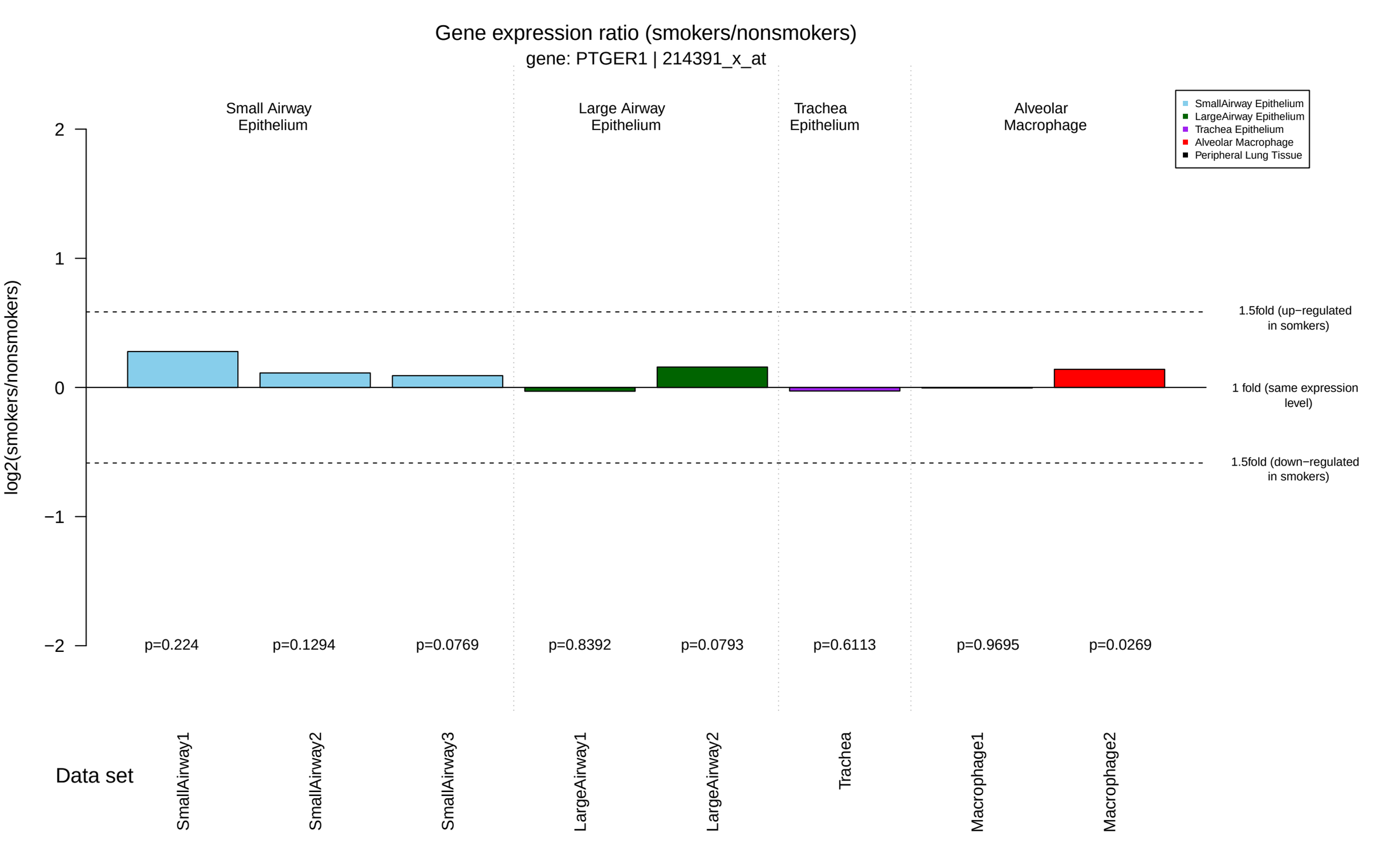
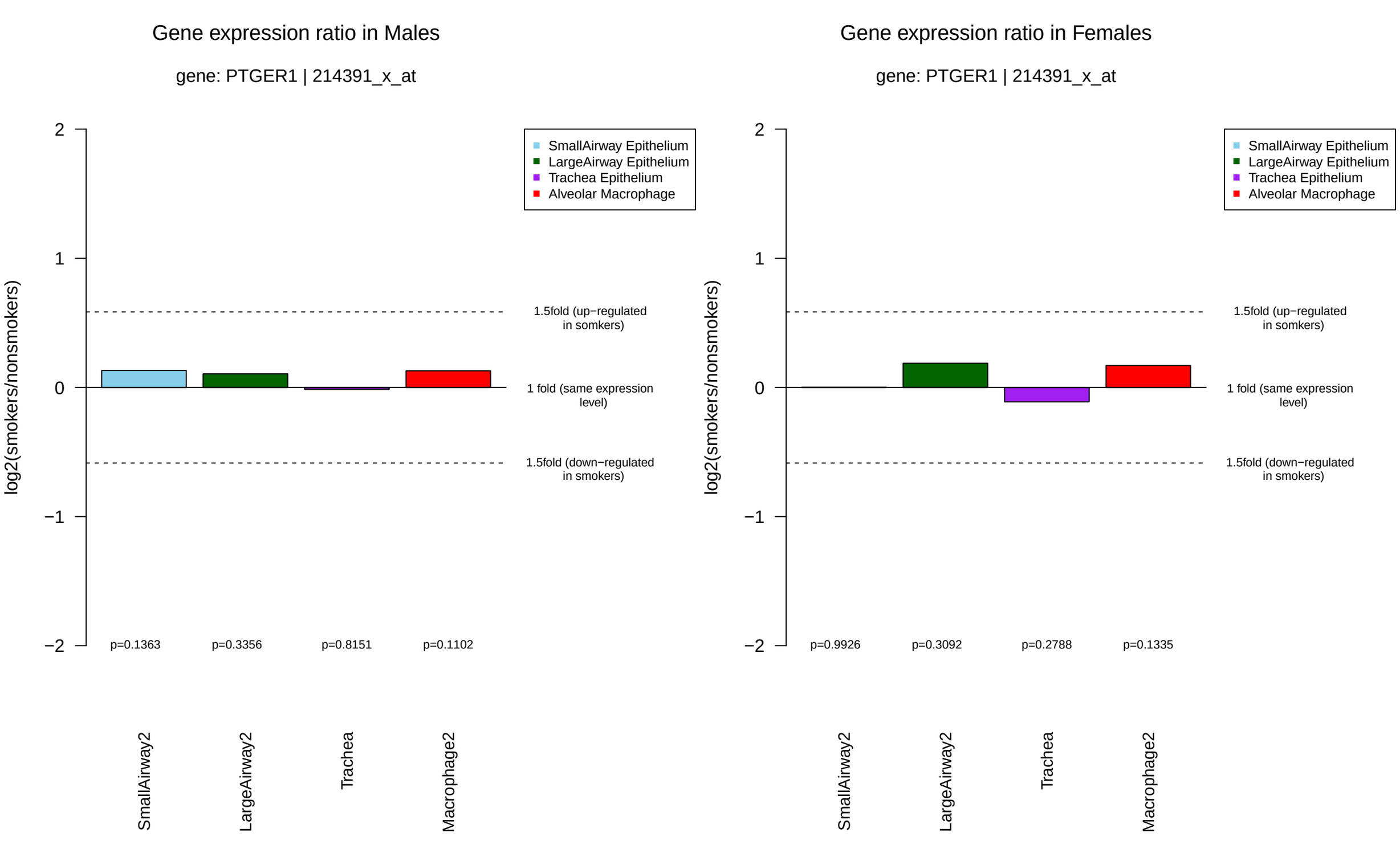
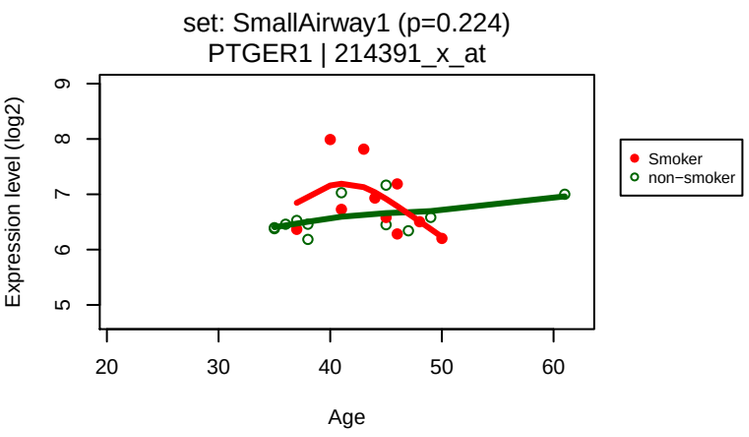
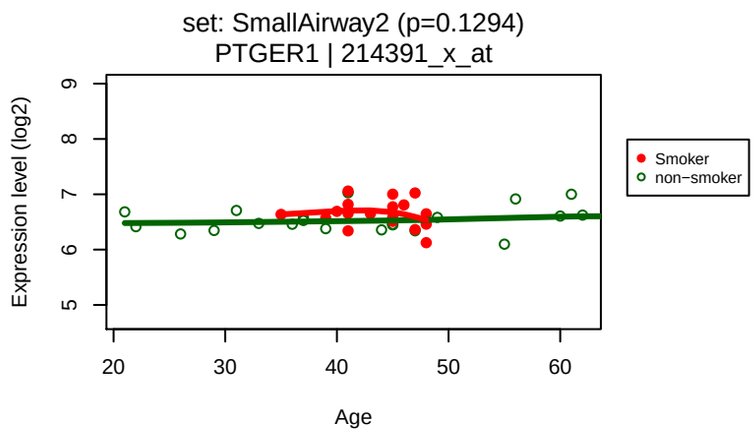
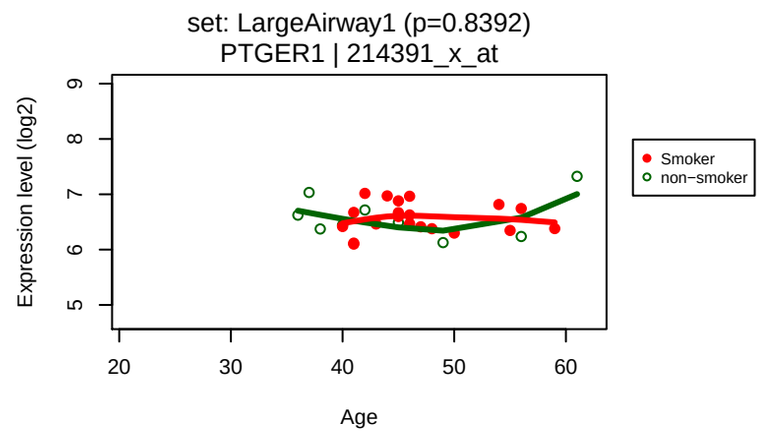

Gene expression of smokers and nonsmokers
gene: PTGER1 | 214391_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.224
p=0.1294
p=0.0769
p=0.8392
p=0.0793
p=0.6113
p=0.9695
p=0.0269
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

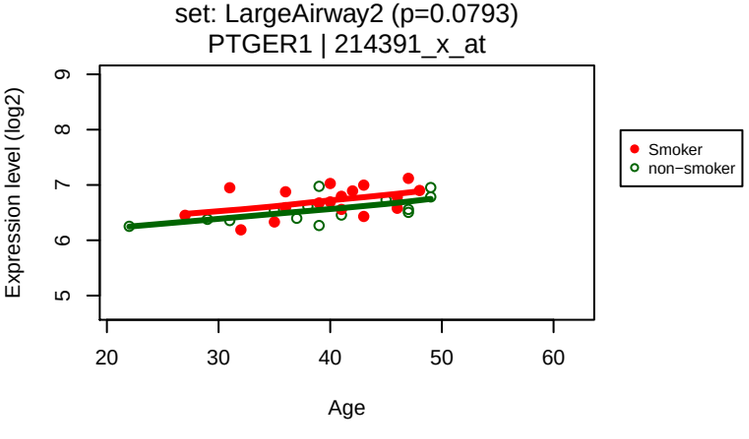

Gene expression ratio (smokers/nonsmokers)
gene: PTGER1 | 214391_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.224
p=0.1294
p=0.0769
p=0.8392
p=0.0793
p=0.6113
p=0.9695
p=0.0269
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

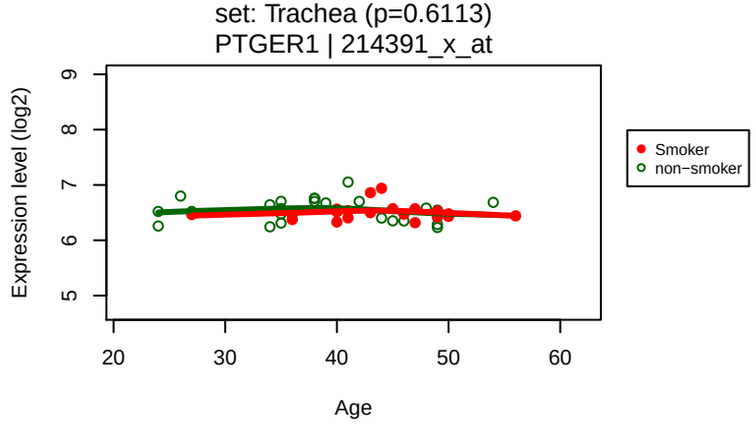

Gene expression ratio in Males
gene: PTGER1 | 214391_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1363
p=0.3356
p=0.8151
p=0.1102
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGER1 | 214391_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9926
p=0.3092
p=0.2788
p=0.1335
SmallAirway2
LargeAirway2
Trachea
Macrophage2

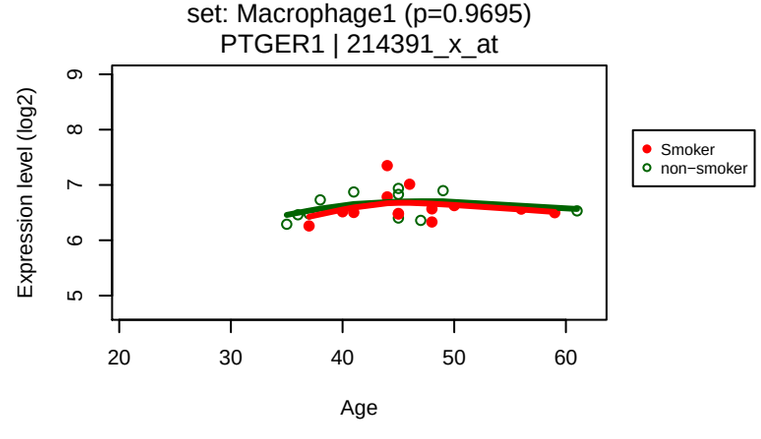

set: SmallAirway1 (p=0.224)
PTGER1 | 214391_x_at
Expression level (log2)
Age
Smoker
non−smoker

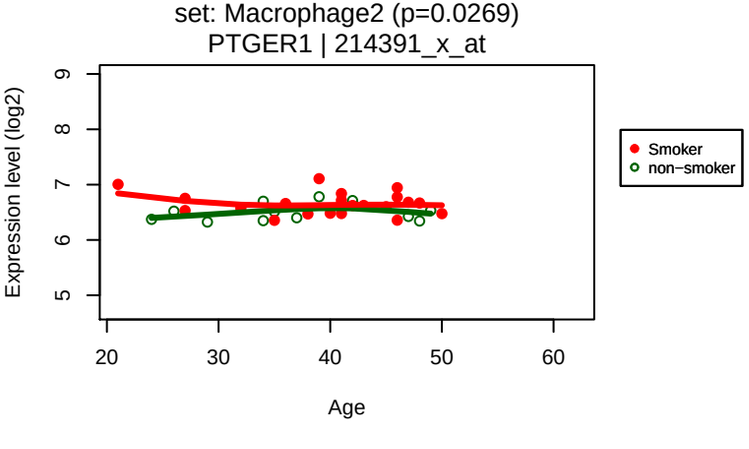

set: SmallAirway2 (p=0.1294)
PTGER1 | 214391_x_at
Expression level (log2)
Age
Smoker
non−smoker

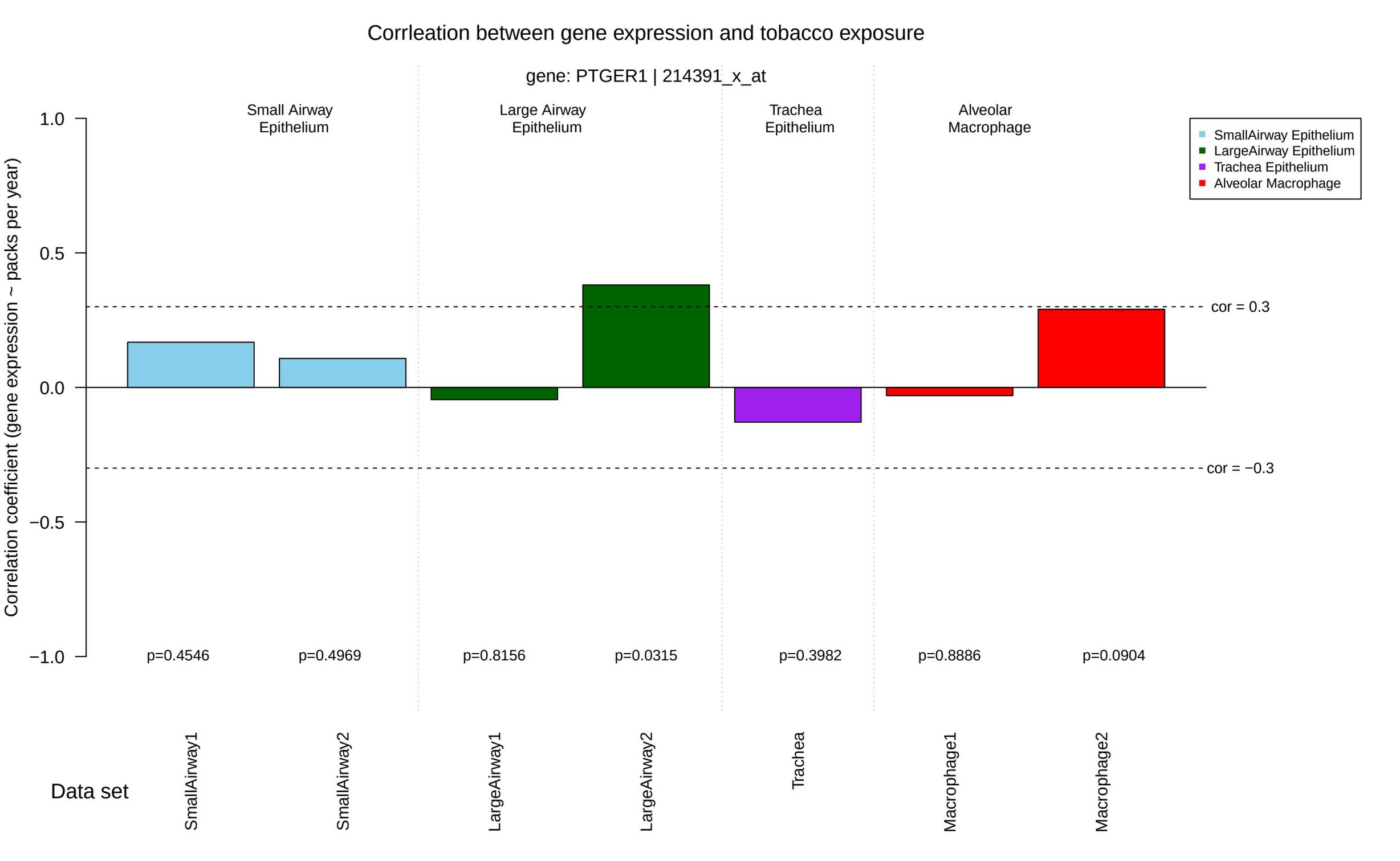

set: LargeAirway1 (p=0.8392)
PTGER1 | 214391_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0793)
PTGER1 | 214391_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6113)
PTGER1 | 214391_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9695)
PTGER1 | 214391_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0269)
PTGER1 | 214391_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGER1 | 214391_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4546
p=0.4969
p=0.8156
p=0.0315
p=0.3982
p=0.8886
p=0.0904
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2